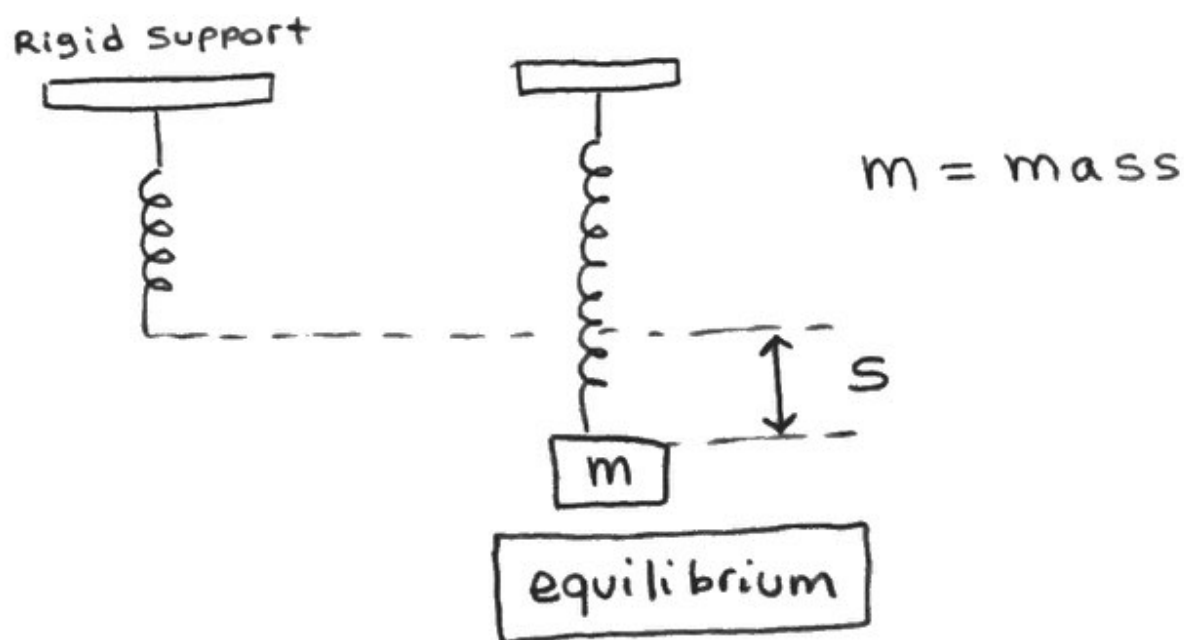


Chapter 5 Modeling with Higher-order DEs

5.1 Linear Models : Initial-Value Problems

Mass Spring systems

Consider a spring attached to a rigid support. If we attach a mass to the spring, gravity will pull the mass downward and the restoring force of the spring will pull the mass upward. When these two forces are equal, the mass is in equilibrium.



The restoring Force of the spring obeys Hooke's Law and it has a magnitude of $F_s = k s$. The constant k is called the spring constant.

The weight of the mass is $F_g = mg$.

At equilibrium, these two forces are equal

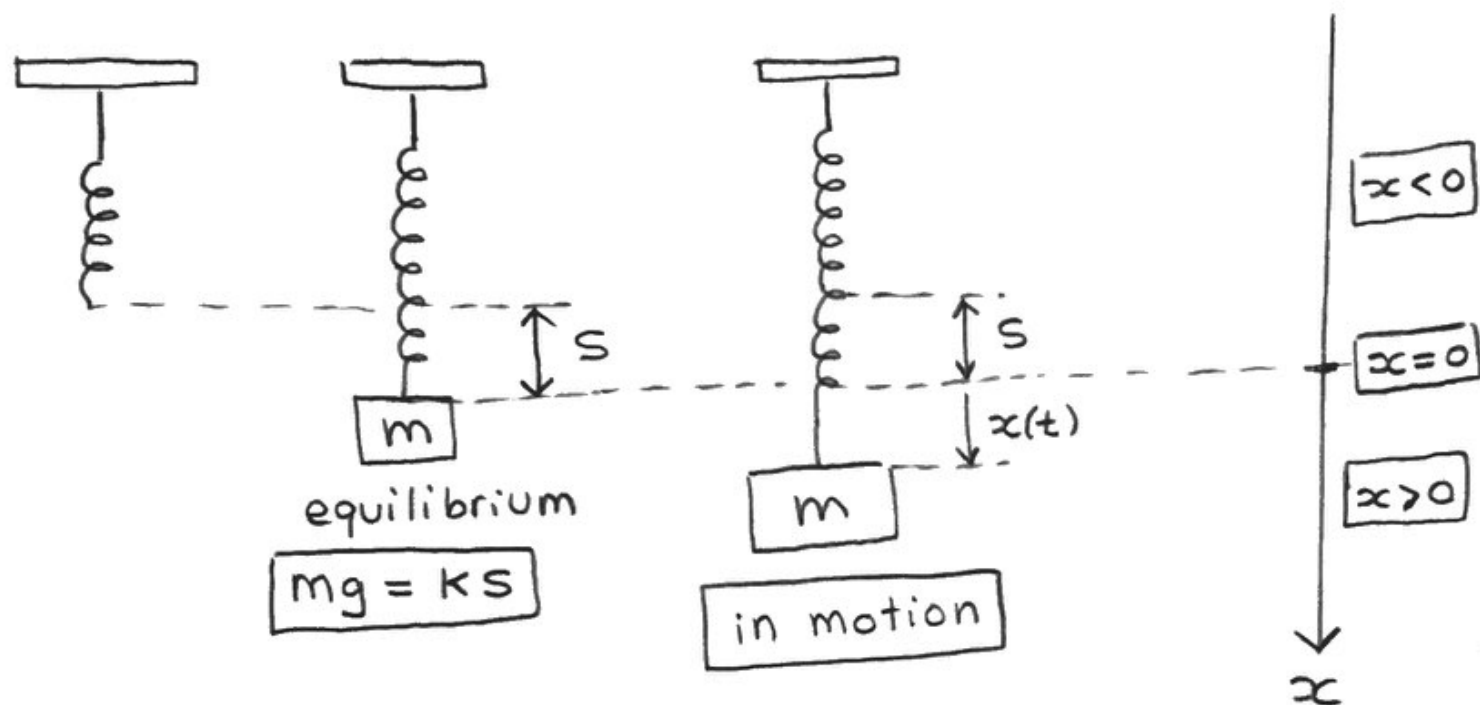
$$\boxed{mg = ks}.$$

If the mass is displaced from equilibrium and is released, it will start to move. Its motion can be modeled by a 2nd order linear DE.

We use Newton's second Law

$$\vec{F} = m \vec{a}$$

to set up the differential equation.



$x(t)$ is the position of the mass at time t .

$x = 0 \rightarrow$ equilibrium

We will use a coordinate system where:

$x > 0$ is below equilibrium

and

$x < 0$ is above equilibrium

Let's assume for now that there are only two forces on the mass: its weight and the restoring force of the spring.

$$\vec{F} = m\vec{a}$$

$$\Rightarrow \underbrace{mg - k(s+x(t))}_{\text{net force}} = \underbrace{m x''(t)}_{ma}$$

$$\Rightarrow \underbrace{mg - ks}_{\text{since } mg = ks} - kx(t) = m x''(t)$$

since $mg = ks$

$$\Rightarrow -kx(t) = m x''(t)$$

$$\Rightarrow \boxed{m x''(t) + kx(t) = 0}$$

By dividing by m , the DE becomes:

$$x''(t) + \frac{k}{m} x(t) = 0$$

or

$$\boxed{x''(t) + \omega^2 x(t) = 0} \quad (*)$$

where : $\omega^2 = \frac{k}{m}$.

To solve $(*)$, we find the roots of :

$$M^2 + \omega^2 = 0 \rightarrow M = \pm i\omega$$

The general solution of $(*)$ is then:

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t).$$

It is often more convenient to write $x(t)$ as

$$\boxed{x(t) = A \sin(\omega t + \phi)}$$

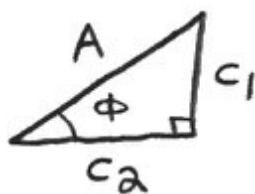
If $x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$, we can find A and ϕ so that:

$$x(t) = A \sin(\omega t + \phi).$$

Trig. identity: $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$$\Rightarrow x(t) = \underbrace{A \sin \phi}_{c_1} \cos(\omega t) + \underbrace{A \cos \phi}_{c_2} \sin(\omega t)$$

$$\Rightarrow \begin{cases} c_1 = A \sin \phi \\ c_2 = A \cos \phi \end{cases}$$



$$A = \sqrt{c_1^2 + c_2^2}$$

$$\tan \phi = \frac{c_1}{c_2}$$



$$\phi = \text{Arctan}\left(\frac{c_1}{c_2}\right)$$



$$\phi = \pi + \text{Arctan}\left(\frac{c_1}{c_2}\right)$$

$$x(t) = A \sin(\omega t + \phi)$$

$$\text{Amplitude} = |A|$$

$$\text{Period: } T = \frac{2\pi}{\omega}$$

ϕ is a phase angle

ex: ① Express $x(t) = 2 \cos(3t) - 5 \sin(3t)$
as $x(t) = A \sin(\omega t + \phi)$.

$$\omega = 3 \quad (\text{By inspection})$$

$$\begin{cases} A \sin \phi = 2 \\ A \cos \phi = -5 \end{cases}$$



$\begin{array}{c} \oplus \\ - \end{array}$	$\begin{array}{c} + \\ - \end{array}$	$\begin{array}{c} \ominus \\ - \end{array}$	$\begin{array}{c} + \\ + \end{array}$
sin		cos	

$$A = \sqrt{4 + 25} = \sqrt{29}$$

$$\phi = \pi + \arctan\left(\frac{2}{-5}\right) \approx 2.761$$

Radians

$$\therefore \boxed{x(t) = \sqrt{29} \sin(3t + 2.761)}$$

ex: (2) A mass of 2 kg is attached to a spring with constant $k = 32 \text{ N/m}$.

If the mass is released from 30 cm below equilibrium with an upward velocity of 0.8 m/s ,

a) Find the position of the mass as a function of time t .

b) Find the Amplitude and Period.

c) Find all times ($t > 0$) when the mass passes through equilibrium.

(i.e., find all $t > 0$ when $x(t) = 0$.)

d) Find the first time when the mass reaches its lowest point.

Let's start from : $m x''(t) + K x(t) = 0$.

$$m = 2 \text{ kg and } K = 32 \text{ N/m}$$

$$\Rightarrow 2 x''(t) + 32 x(t) = 0$$

$$\Rightarrow x''(t) + 16 x(t) = 0$$

General solution : $x(t) = c_1 \cos(4t) + c_2 \sin(4t)$.

$x(t)$ is in meters.

$$30 \text{ cm} = 0.3 \text{ m}$$

The initial conditions are :

$$\begin{cases} x(0) = + 0.3 \text{ m} & (+ \text{ since } \underline{\text{below}}) \\ x'(0) = - 0.8 \text{ m/s} & (- \text{ since } \underline{\text{upward}}) \end{cases}$$

$$x'(t) = -4c_1 \sin(4t) + 4c_2 \cos(4t)$$

$$x(0) = 0.3 \rightarrow c_1 + 0 = 0.3 \rightarrow c_1 = 0.3$$

$$x'(0) = -0.8 \rightarrow 0 + 4c_2 = -0.8 \rightarrow c_2 = -0.2$$

a) $x(t) = 0.3 \cos(4t) - 0.2 \sin(4t)$

b) Let's express $x(t)$ as

$$x(t) = A \sin(\omega t + \phi).$$

$$x(t) = 0.3 \cos(4t) - 0.2 \sin(4t)$$

$$\begin{cases} A \sin \phi = 0.3 \\ A \cos \phi = -0.2 \end{cases}$$



$$\Rightarrow \begin{cases} A = \sqrt{(0.3)^2 + (-0.2)^2} = 0.36 \\ \phi = \pi + \arctan\left(\frac{0.3}{-0.2}\right) = 2.1588 \end{cases}$$

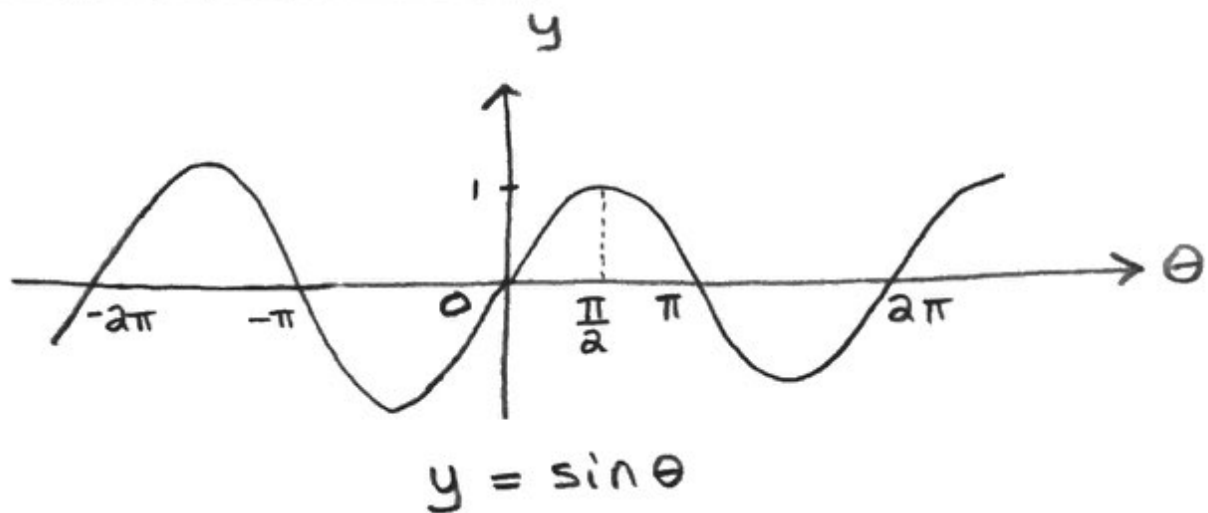
$$x(t) = 0.36 \sin(4t + 2.1588)$$

Amplitude : 0.36 m or 36 cm

Period : $T = \frac{2\pi}{\omega} = \frac{2\pi}{4}$ seconds

$$T = \frac{\pi}{2} \text{ seconds}$$

c)



$$\sin \theta = 0 \rightarrow \theta = n\pi \text{ for } n=0, \pm 1, \pm 2, \dots$$

$$x(t) = 0 \rightarrow \sin(4t + 2.1588) = 0$$

$$\rightarrow 4t + 2.1588 = n\pi$$

$$\rightarrow \boxed{t = \frac{n\pi - 2.1588}{4}}$$

$$t > 0 \text{ for } n = 1, 2, 3, 4, \dots$$

d) The lowest point occurs when :

$$x(t) = +A \quad (\text{below equilibrium})$$

the highest point would be when :

$$x(t) = -A \quad (\text{above equilibrium})$$

$$\sin \theta = +1 \text{ for } \theta = \frac{\pi}{2} + 2n\pi$$

$$n = 0, \pm 1, \pm 2, \dots$$

$$x(t) = A \rightarrow \sin(\omega t + \phi) = +1$$

$$\text{For our case, } \sin(4t + 2.1588) = 1$$

$$\Rightarrow 4t + 2.1588 = \frac{\pi}{2} + 2n\pi$$

$$\Rightarrow t = \frac{\frac{\pi}{2} + 2n\pi - 2.1588}{4}$$

$$n=0 \rightarrow t = \frac{\frac{\pi}{2} - 2.1588}{4} < 0$$

$$n=1 \rightarrow t = \frac{\frac{\pi}{2} + 2\pi - 2.1588}{4} = 1.42 \text{ s}$$

The first time when the mass reaches its lowest point is

$$t = 1.42 \text{ seconds}$$

Damped motion

Let's assume that there is a damping force that is proportional to the velocity of the mass and acts in the opposite direction.

Damping Force : $F_d = -\beta v$

for a constant $\beta > 0$ called the damping constant.

$$\underbrace{mg - k(s+x(t)) - \beta x'(t)}_{\text{net force}} = m x''(t)$$

$$\Rightarrow \underbrace{mg - ks}_{=0} - kx(t) - \beta x'(t) = m x''(t)$$

$$\Rightarrow \boxed{m x''(t) + \beta x'(t) + kx(t) = 0}$$

The characteristic equation of the DE is :

$$m \lambda^2 + \beta \lambda + k = 0.$$

(we use the variable λ since m is used for mass)

The roots are :

$$\lambda = \frac{-\beta \pm \sqrt{\beta^2 - 4mk}}{2m}$$

We will look at three cases.

case ① $\beta^2 > 4mk$ (overdamped)

Case ② $\beta^2 = 4mk$ (critically damped)

Case ③ $\beta^2 < 4mk$ (underdamped)

Case ①

Overdamped

$$\beta^2 > 4mk$$

$$\begin{cases} \lambda_1 = \frac{-\beta + \sqrt{\beta^2 - 4mk}}{2m} < 0 \\ \lambda_2 = \frac{-\beta - \sqrt{\beta^2 - 4mk}}{2m} < 0 \end{cases}$$

The general solution is :

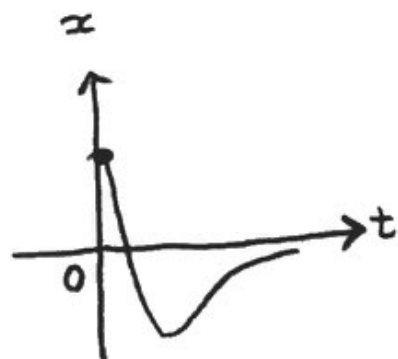
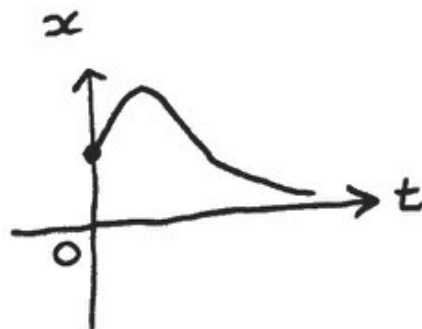
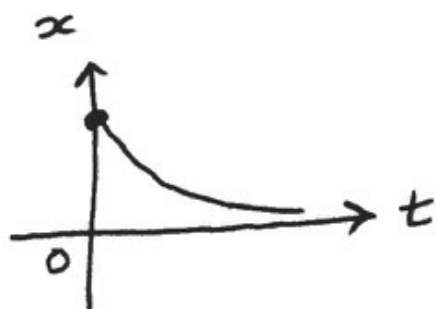
$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

Since $\lambda_1 < 0$ and $\lambda_2 < 0$, $x(t) \rightarrow 0$

as $t \rightarrow +\infty$. It decays to zero over time.

Overdamped systems do not oscillate.

Typical graphs look like :



Case (a) critically damped $\beta^2 = 4mk$

There is only one root $\lambda = -\beta/2m$.

The general solution is:

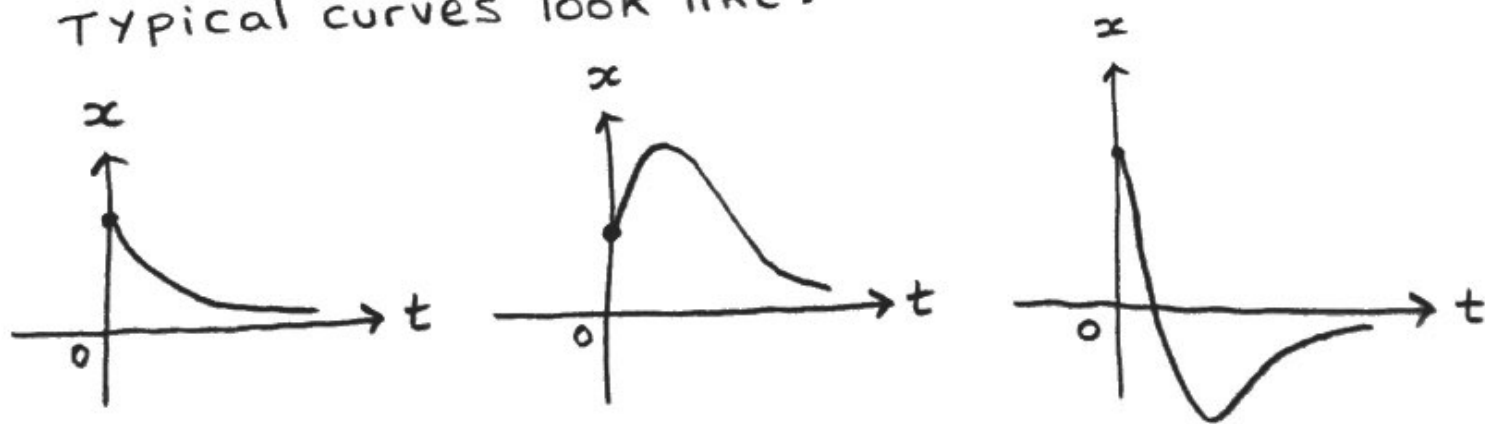
$$x(t) = (c_1 + c_2 t) e^{-\beta t/2m}.$$

The motion is very similar to an overdamped system. There is no oscillation.

If the damping is reduced by any amount, the system would start to oscillate.

We still have $x(t) \rightarrow 0$ as $t \rightarrow +\infty$.

Typical curves look like:



In practice it is almost impossible to have a mass spring system with β^2 exactly equal to $4mk$. Critically damped is simply the theoretical boundary between overdamped and underdamped.

Case ③

Underdamped

$$\beta^2 < 4mk$$

There are complex roots:

$$\lambda = \frac{-\beta}{2m} \pm \frac{\sqrt{4mk - \beta^2}}{2m} i$$

$$\text{Let } \omega = \frac{\sqrt{4mk - \beta^2}}{2m}$$

$$\Rightarrow \lambda = \frac{-\beta}{2m} \pm i\omega$$

The general solution is

$$x(t) = e^{-\frac{\beta t}{2m}} \left[c_1 \cos(\omega t) + c_2 \sin(\omega t) \right]$$

We can express it as

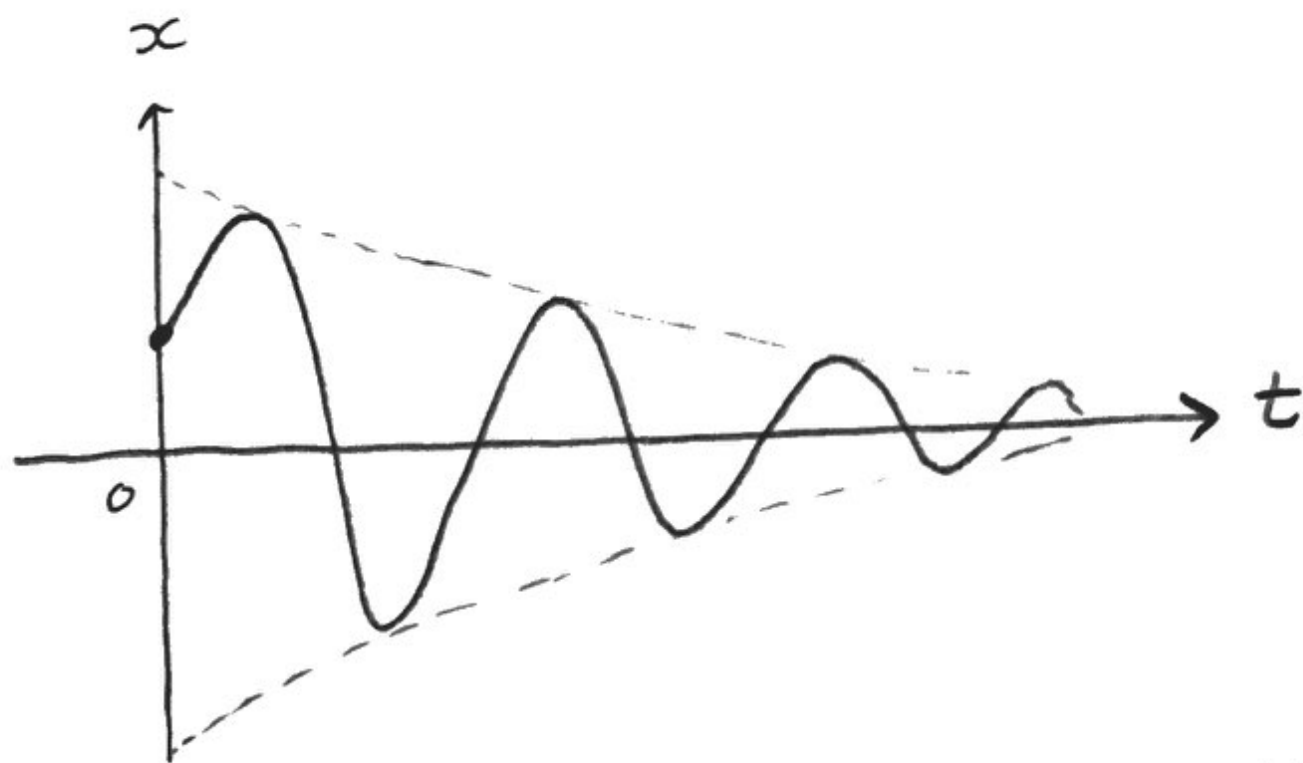
$$x(t) = A e^{-\frac{\beta t}{2m}} \sin(\omega t + \phi)$$

$$x(t) = \boxed{A e^{-\beta t / 2m}} \sin(\omega t + \phi)$$

↓
"Decaying Amplitude"

as $t \rightarrow +\infty$, we have: $x(t) \rightarrow 0$.

A typical underdamped graph is :



The system oscillates with an Amplitude that decays to zero over time.

Note : If $\beta = 0$ (No damping)

then $e^{-\beta t/2m} = 1$ so that

$$x(t) = A \sin(\omega t + \phi).$$

If $\beta > 0$, in all three cases

we get that :

$$\lim_{t \rightarrow +\infty} x(t) = 0.$$

The system always approaches equilibrium over time when there is damping.

ex: ③ A 1 kg mass is attached to a spring of constant 10 N/m.

Assume that the damping force is equal to twice the instantaneous velocity.

Find the position of the mass if it is released from equilibrium with a downward velocity of 2 m/s.

We start from :

$$m x''(t) + \beta x'(t) + k x(t) = 0.$$

$$\left\{ \begin{array}{l} m = 1 \text{ kg} \\ \beta = 2 \frac{\text{N} \cdot \text{s}}{\text{m}} \\ k = 10 \text{ N/m} \end{array} \right. \quad (\text{twice the velocity})$$

$$x''(t) + 2x'(t) + 10x(t) = 0$$

initial conditions

$$\begin{cases} x(0) = 0 & (\text{equilibrium}) \\ x'(0) = +2 \text{ m/s} & (+ \text{ since downward}) \end{cases}$$

To solve the DE, we find the roots of

$$\lambda^2 + 2\lambda + 10 = 0$$

$$\Rightarrow \lambda = -1 \pm 3i$$

$$\Rightarrow x(t) = e^{-t} (c_1 \cos(3t) + c_2 \sin(3t))$$

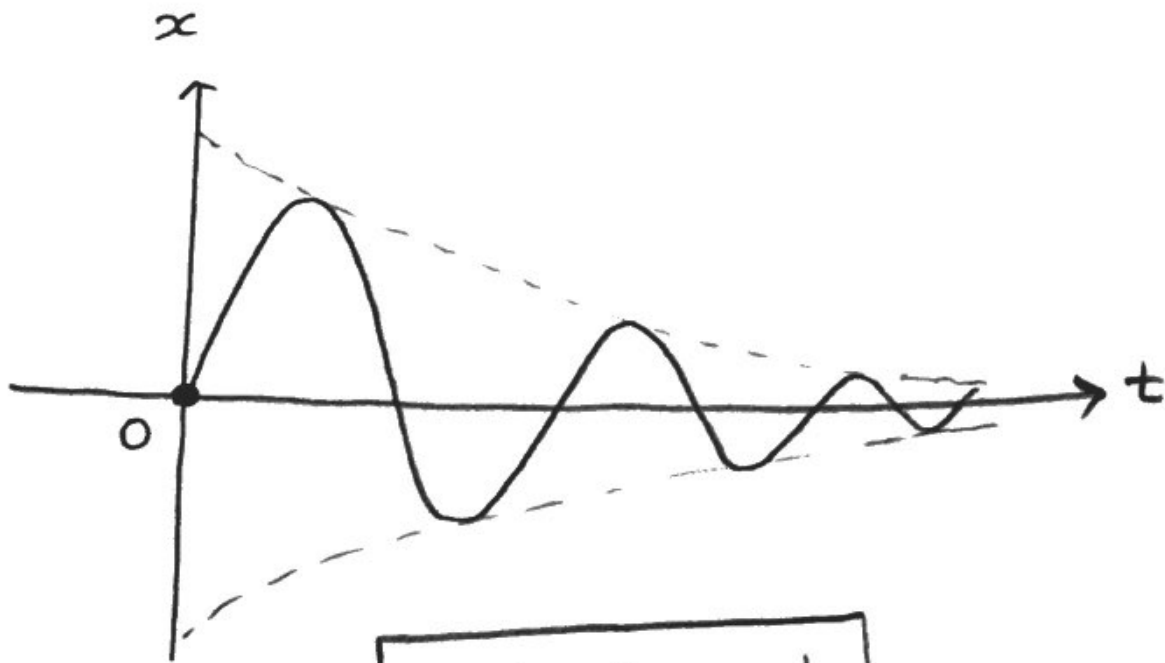
$$x(0) = 0 \rightarrow e^0 (c_1 + 0) = 0 \rightarrow c_1 = 0$$

$$\Rightarrow x(t) = c_2 e^{-t} \sin(3t)$$

$$x'(t) = -c_2 e^{-t} \sin(3t) + 3c_2 e^{-t} \cos(3t)$$

$$x'(0) = 2 \rightarrow 0 + 3c_2 = 2 \rightarrow c_2 = 2/3$$

$$\therefore x(t) = \frac{2}{3} e^{-t} \sin(3t)$$



underdamped